Root-library validation readiness

Source-only preparation under T-1636. The owner dollar waiver is accepted; it does not change the worker ceiling, deadline, memory, CPU, one-payload concurrency, retry limit, input custody or tests. No payload ran in this task.

The existing test sequence is concrete: cargo test --release --lib --locked --offline --no-run --message-format=json; discover exactly one crypto_lib test executable; list tests; run the same four exact named tests, including actual_some_none_known_answer_and_diagnostic_ledger; then cargo test --release --lib --locked --offline -- --test-threads=1. Existing wrapper total limit remains 1800 seconds, 2 CPU, 16 GiB, no swap, one Cargo job and one test thread.

All 259 archived crate paths and all 86 tool-input paths exist. Cargo and rustc bytes match the old recorded hashes. The old release cache directory exists. Full archive/cache/sysroot identity has not been rechecked; existence is not custody qualification. The historical cache receipt records 2,384,395,266 cache bytes and 261,870,472 expanded vendor bytes; these are historical counts.

Pinned commit 29861c9a55f9ae5a33e1e09b3866aa397fbde605 retains the exact Cargo.toml, lockfile and canonical pencil source in the Git object database. The candidate patch and fixture survive unchanged. The former /var/tmp source/vendor/target workspace is absent. A fresh private source tree, vendor extraction and selected cache copy are needed; materialized-inputs.json must bind their new paths and complete default-feature/include closure. No broad shared writable cache or network fallback is needed by design.

Filesystem readback shows /home/ec2-user and /srv/experiments on migrated NFS4; its apparent exabyte free-space value is not a physical capacity guarantee. /var/tmp is XFS with about 90 GiB available at readback. Native logs remain in the durable home path as specified by v2, with fixed byte thresholds. Workspace/cache can use a fresh /var/tmp path, separate from durable receipts.

The minimal versioned amendment changes only the admission helper: bind the explicit owner instruction instead of requesting dollar estimates/allocation. It preserves exact tests, all recipe limits, publication bindings, leases and no-foreign-kill behavior. The v2 files are unchanged. The new helper and patch are unexecuted and require source review before use; root must populate the hash-bound owner instruction receipt and current campaign debit.

Concrete remaining steps: root selects the existing pinned revision or explicitly rebinds a newer one; prospectively bounded materialization recreates source/vendor/cache and hashes; publish those exact input/recipe/admission bindings; qualify wrapper containment and output persistence and run the existing four controls plus full suite after the current norm payload is terminal. A successful compile alone cannot substitute for the full suite. No unmeasured cost is assigned zero.

Sources: durable-v2 PLAN.md, recipe-proposal.json and durable-worker.py; historical source-materialization.json, root-check-tool-inputs.json, registry-archives.json and cache-preparation-v2-receipt.json. These existing recipes establish the preparation requirements; no new mathematical mechanism or literature claim is introduced.